

CoRoT-2b

R-0066

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_250518 · created 2026-07-15T15:49:28+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460813.8619 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.018
Transit depth (Rp/Rs)^2	2.5 +/- 0.57 [%]
Orbital Inclination (inc)	83.94 +/- 0.81
Airmass coefficient 1 (a1)	1.067 +/- 0.014
Airmass coefficient 2 (a2)	-0.055 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.3 %
Best Comparison Star	#3 - [539, 401]
Optimal Aperture	3.37
Optimal Annulus	12.31
Transit Duration (day)	0.0637 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.158 $\pm$ 0.018 vs 0.1667 (deviation 0.36 σ)
Duration fit vs published	0.0637 d vs 0.095 d (deviation 5.44 σ)
Mid-time O-C	-3.2 min
Depth SNR	6.6
Residual scatter	1.3 %

Light curve

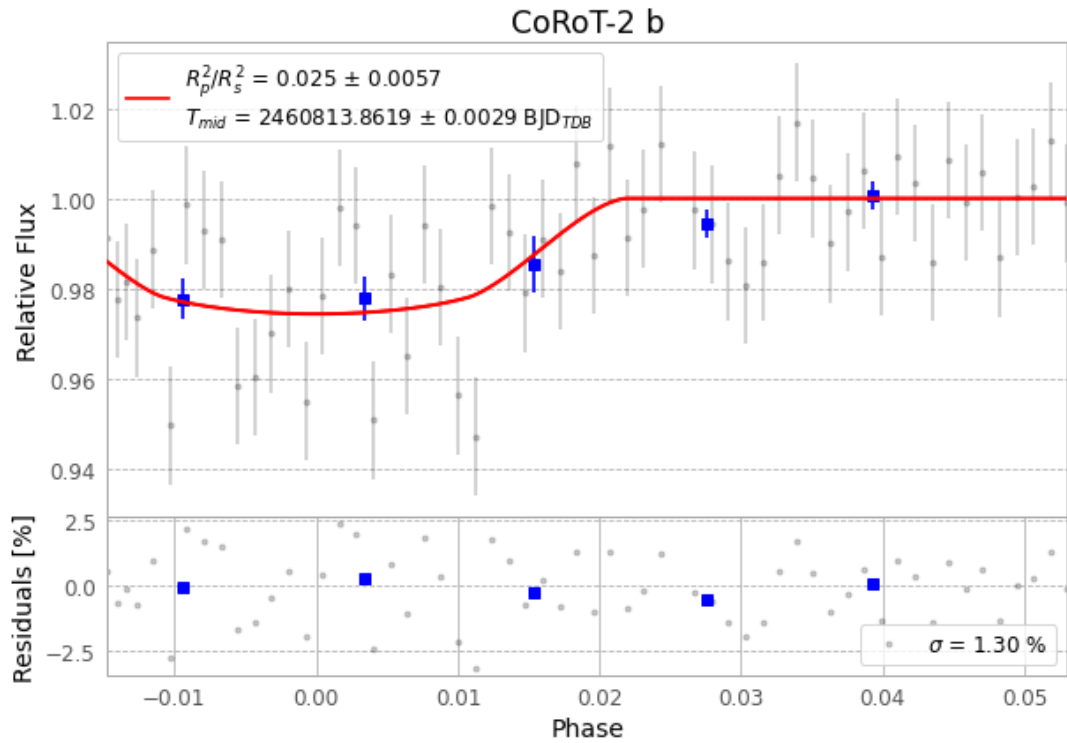


Figure 1 — FinalLightCurve_CoRoT-2 b_2025-05-18.png (SHA-256 2acea0a0c4404265...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [333, 197], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 361fb51ecdacd6c9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

59 FITS frames (39 MB), CoRoT-2250518075805.FITS → CoRoT-2250518104816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-250518.log (e3505f586503...), FinalLightCurve_CoRoT-2 b_2025-05-18.png (2acea0a0c440...), FinalLightCurve_CoRoT-2 b_2025-05-18.csv (3910242123d7...)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.